

Seven Ways to Effectively Use Today's Feature-Heavy Scopes



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Since its inception, the oscilloscope is one test and measurement product that has seen a lot of transformations to deal with the needs of new technological advancements. "Recent advancements in analogue and digital electronics with growth in telecom, medical electronics, automotive and other sectors have brought revolutionary changes in the scope," says Vivek Mantri, country manager, industrial segment, Scientech Technologies Pvt Ltd. Requirements have elevated measurement parameters from MHz to GHz, when it comes to bandwidth of the scope. He adds, "Right from an engineering student analysing his basic astable multivibrator to engineers testing and verifying complex designs, the oscilloscope is a useful tool."

But, there are often some features and methods that are unused or missed-out while testing designs. In this attempt, we try to cover some important functions of the oscilloscope that would help you effectively use it and efficiently test your designs.

Use memory depth feature for finer signal analysis

While considering an oscilloscope, the obvious parameters that you look for are bandwidth, acquisition rate and sampling rate, but one feature that is often overlooked is memory depth. To achieve highly-accurate calculations, memory depth is said to be one of the top features of oscilloscopes today. This feature allows better monitoring, triggering and analysis of signals, especially applicable on those signals with high-frequency components. Obtaining extended time periods with the capability to zoom into the signal, without loss of data, allows an engineer to verify the hardware design in a swift and proper manner.

Some digital oscilloscopes provide the ability of encapsulating waveforms in long-memory mode. For standard oscilloscopes, memory depth ranges from 1kpt to 16kpt (pt is point). With the long-memory feature, oscilloscopes can capture complex signals in great detail over extended time periods. This allows test engineers to inspect effects of high frequency within the captured waveform.

